Wi-Fi Sensing for McDonald's Self Service Kiosks

by Damla Bortucen, Alex Pison, Teodora Lentu, James Glover

Table of contents

1. Introduction
2. Wheelchair users
3. Visually impaired users
4. Neurodivergent users
5. Users with limited hand mobility or motor impairments
6. Conclusion
7. Bibliography

1. Introduction

Since their introduction in 2015, McDonald's self-order kiosks have quickly become the preferred method of making an in-store order. A 2023 study showed that 73% of customers were familiar with the usage of self-service kiosks, compared to 31% ordering by mobile or 24% by delivery [1].

Becoming the standard and replacing cashiers in stores all over the UK, customers no longer get a choice in their ordering method. While the average customer enjoys this method of ordering, for some demographics it's turned into an uncomfortable experience. The current McDonald's experience **excludes** the following customers: **wheelchair users, visually impaired users, neurodivergent users and those with limited hand mobility or motor impairments.**

Every customer should also be able to rely on a smooth and accessible in-store experience. Hence this report focuses on the use of Wi-Fi sensing to enhance the self-service kiosk experience of people with disabilities.

2. Wheelchair users

The Problem

Based on our research, wheelchair users cannot reach the top of the display, compromising their satisfaction with the service. The top of the kiosk display is 1.79 meters high. This is not a height comfortably accessible by a wheelchair user. The ideal reach zone should start at 450mm off the ground and then be roughly 840mm tall [2], as shown in Figure 1. This makes 510mm of kiosk screen space unreachable.

According to a 2022 Motability Foundation study, estimates for wheelchair usage in England range from 688,000 to 860,000, or up to 1.37 million if mobility scooters were included [3]. This amounts to approximately 2% of the English population, whose needs are being overlooked [4].

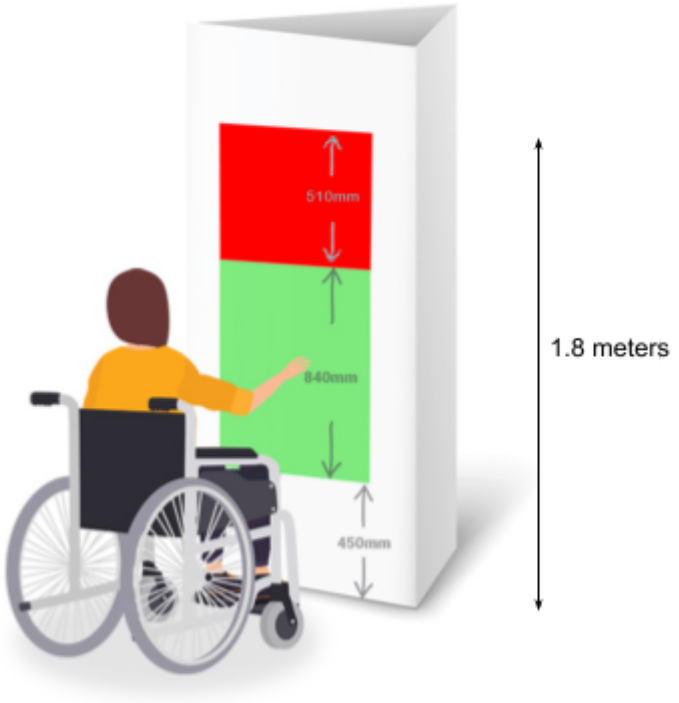

Figure 1: Ideal reach zone of a wheelchair user (green) vs unreachable kiosk space (red) [2]

The Solution

Automatic resizing of the McDonald's kiosk's UI based on the user using **Wi-Fi sensing**. Using machine learning and state of the art Wi-Fi 7 technology, we are able to detect what kind of user is interacting with the kiosk **without needing to install any additional hardware**.

If a wheelchair user approaches the kiosk, the UI resizes, allowing the person on wheelchair to order without any restrictions.

As shown in Figure 2, the wheelchair user would approach the kiosk, the kiosk would automatically detect that and resize accordingly.

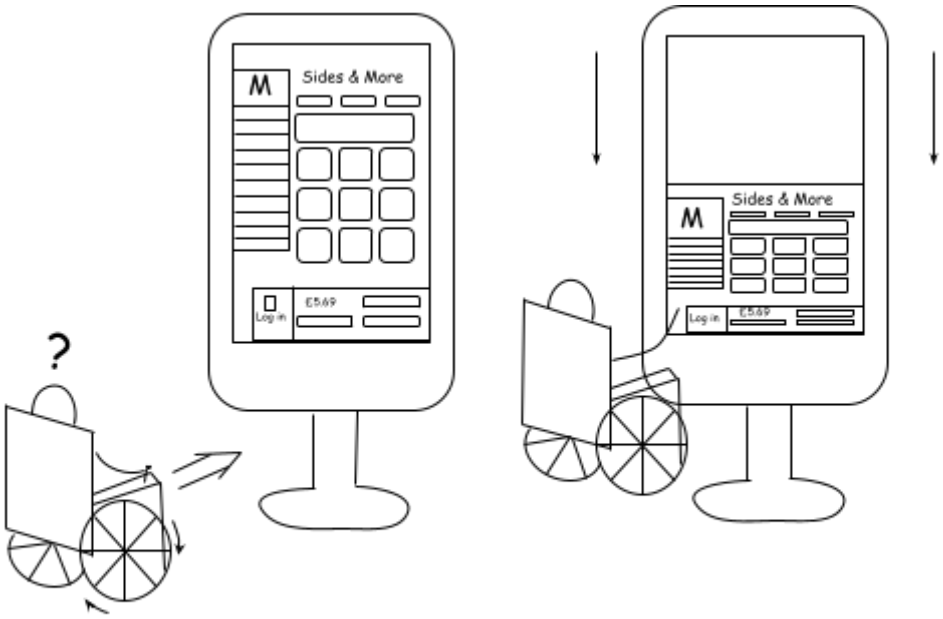

Figure 2: UI Resizing Diagram for Wheelchair Users

Why is this better?

By using an automatic trigger,

- We remove the need to use the dedicated button to trigger UI resizing. People may not know about the existence of this button in the first place.
- We would make the McDonald's customers on wheelchairs feel more welcomed and included in the restaurants.

3. <u>Visually impaired users</u>

The Problem
According to a government survey done by the Korean Ministry of Health and Welfare in 2024, it was found that 7 in 10 visually impaired people preferred ordering from a staff member in comparison to kiosks. Furthermore, the most challenging type of kiosks were named to be self-ordering machines, such as the ones used in McDonalds, by 80.1 percent of survey takers - a group comprised of the visually impaired, blind and wheelchair users. [5]

The Solution
We propose tailoring the kiosk UI for the visually impaired by removing the pictures and instead making the text bigger and high-contrast, and so easier to read. We also suggest decluttering the view by removing side panels and instead making the menu screen separate, as can be seen in Figure 3.

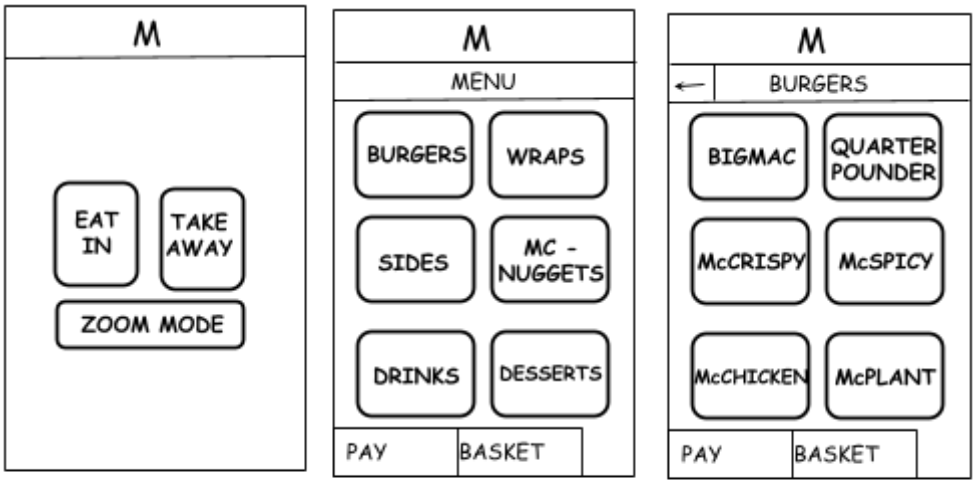

Figure 3: UI Prototype for Visually Impaired Users

Why is this better?
This design directly addresses the core accessibility challenges faced by visually impaired users.

- Removing images and enlarging high-contrast text, the interface becomes much easier to read and navigate, even for users with partial sight.
- Simplifying the layout and decluttering the screen reduces cognitive and visual load, helping users focus on one clear task at a time.
- Separating the menu into distinct screens instead of using side panels ensures that information is presented in a linear and predictable way.

4. <u>Neurodivergent users</u>

The Problem
Neurodivergent people have trouble staying focused while placing an order using the kiosks, because they get overwhelmed by the multitude of options and flashy displays. Having many

options on the screen at once can contribute to **decision fatigue**, decreasing the ease with which they can order at self-serve kiosks [6].

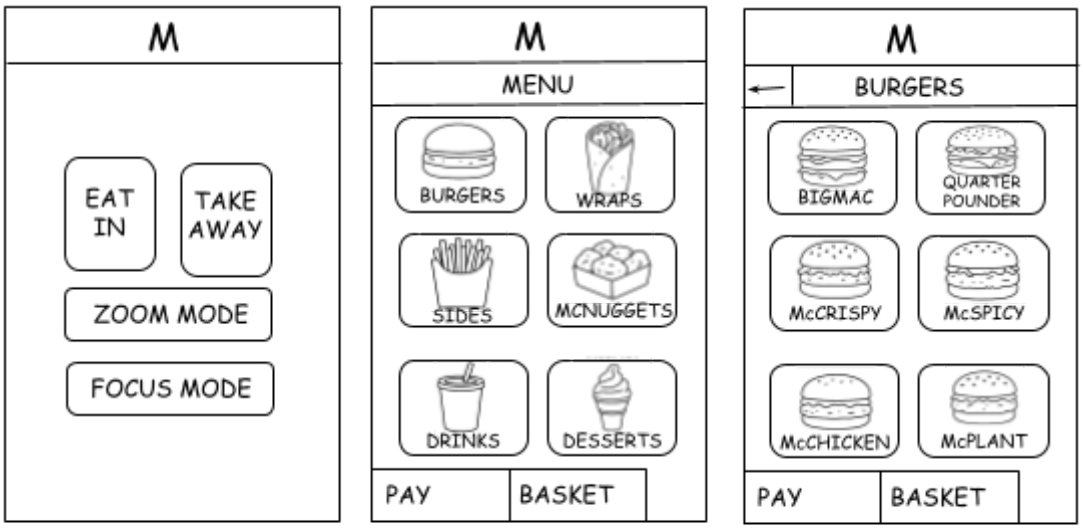

Figure 4: Focus Mode UI for Neurodivergent Users

The Solution

We propose simplifying the kiosk layout by removing side panels, pop-ups, and non-essential visual elements. Furthermore, the interface will use a black-and-white color scheme for most components, keeping only the menu item images in color to help guide attention to the most important parts of the screen.

Why is it better?

This design reduces sensory overload and decision fatigue, which are common challenges for neurodivergent users. Our approach supports clearer navigation, sustained focus, and reduced anxiety, making the ordering process more comfortable and accessible for neurodivergent users. Furthermore, it aligns with inclusive design principles that emphasize simplicity, predictability, and low-stimulation environments.

5. <u>Users with limited hand mobility or motor impairments</u>

The Problem

For individuals with limited hand mobility or motor impairments, such as those with cerebral palsy, arthritis, or tremors, touchscreens can be a significant barrier. These users may struggle to maintain the fine motor control needed to press small buttons or to interact with screens that require extended reach.

McDonald's kiosks interfaces currently feature small densely packed buttons, some which are placed high on the screen. Due to this, these types of users frequently experience high error rates and longer order completion time [7].

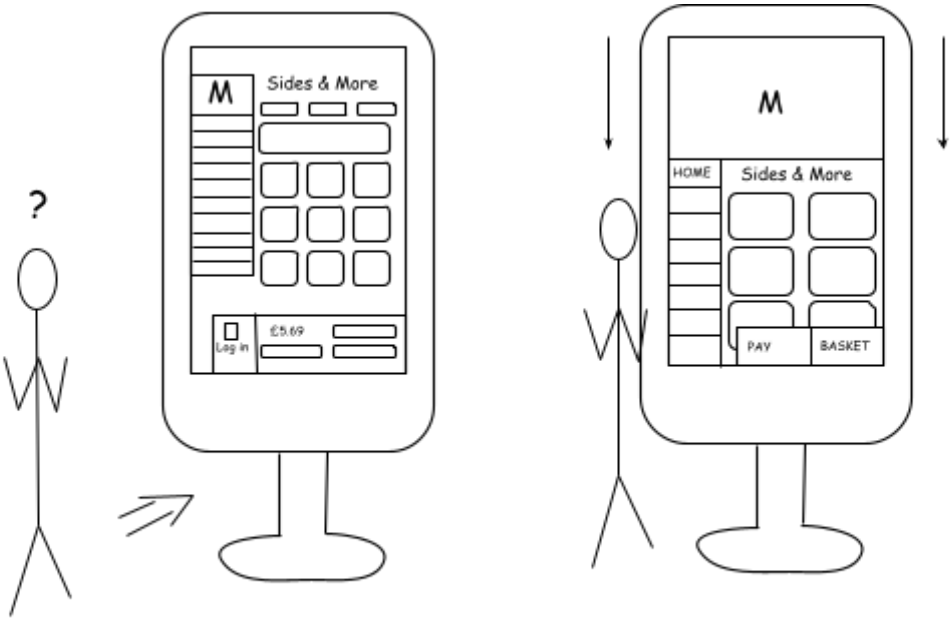

Figure 5: UI Prototype for Impaired Mobility (increased icon size, resized UI)

The solution

We propose redesigning the kiosk with larger buttons and a lowered UI in order to account for the user's disabilities. In addition, we would increase the expected completion time so that the user is not interrupted during their order.

Why is it better?

These changes would make kiosks more inclusive by reducing the precision and speed required to interact with the screen. Research indicates that performance for users with motor impairments continue to benefit from even larger targets [7]. These design improvements would enable users with limited dexterity to order independently, increasing both accessibility and customer satisfaction.

6. Conclusion

Through outlining the short-comings of the McDonald's self-serve kiosks, this report has discussed various ways of addressing the accessibility needs of user demographics that are currently overlooked. The following solutions aim to make the kiosks more inclusive and user-friendly for everyone:

- Wheelchair users - automatic detection for UI resizing
- The visually impaired - high-contrast zoomed in UI
- The neurodivergent - simplification of UI by removing clutter
- People with motor impairments - larger buttons, resized UI, and extended input time

Overall, our proposed solutions aim to make McDonald's a more welcoming and inclusive place for all customers.

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